Venkata Lakshmi Parimala Pasupuleti

Data Analysis · SQL · Python · Tableau · Power BI · Statistical Analysis · Data Quality Management · Data Governance

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PROFESSIONAL SUMMARY

Senior Data Analyst with 8+ years of experience translating complex business requirements into **data-driven solutions** using SQL, Python, Tableau, and Power BI. At Entrata (PropTech SaaS, 100K+ MAU), owned end-to-end **data quality management** across 200+ SQL queries and large-scale financial pipelines — data profiling, cleansing, monitoring, and regulatory compliance on 1.6M+ records. Applied **statistical techniques** (regression, Isolation Forest, Random Forest) across multi-source datasets to surface operational insights that drove strategy. Designed **Tableau leadership dashboards** improving stakeholder comprehension by 40%; mentored junior analysts in data analysis methods, visualization best practices, and quality standards. M.S. Data Analytics Engineering, George Mason University (GPA 3.87, May 2026).

CORE COMPETENCIES

Data Analysis & Exploratory Analysis | SQL & Distributed Data Stores | Data Quality Management | Data Profiling & Data Cleansing | Data Governance & Metadata Standards | Regulatory Compliance & Data Privacy | Statistical Analysis & Hypothesis Testing | Tableau & Power BI Dashboards | Python (Pandas, NumPy, scikit-learn) | Apache Spark / PySpark | Business Intelligence & KPI Metrics | Cross-Functional Stakeholder Collaboration | Mentoring Junior Analysts | eCommerce & Operational Analytics

EXPERIENCE

George Mason University — Research Associate / Teaching Assistant

Aug 2024 – May 2026

Research Associate · Statistical Analysis · Data Visualization · Data Quality

- Achieved $R^2 = 0.977$ regression accuracy across 190+ countries by building multi-source **statistical modeling** pipelines (Python, R) on WHO and UK Road Safety datasets — rigorous **exploratory data analysis**, hypothesis testing, and data validation driving research strategy recommendations for faculty stakeholders.
- Applied **Isolation Forest anomaly detection** across 3 research pipelines to automate **data quality** checks — surfaced data integrity issues that manual review missed; improved downstream model reliability across all active pipelines.
- Designed **Tableau and ggplot2 leadership dashboards** visualizing KPI metrics across two research cohorts — improved stakeholder comprehension by 40%; performed **data profiling** and exploratory analysis to drive actionable research recommendations.
- Mentored 600+ students in Python, SQL, ER modeling, data validation, and data analysis methods across MIS 303; provided one-on-one debugging support and rubric-aligned feedback.

Entrata (PropTech / Real Estate SaaS)

Sep 2018 – Aug 2024

Senior Software Engineer — Data Analysis · SQL Optimization · Data Quality · Regulatory Compliance

- **Optimized 200+ SQL queries** across all client environments (30% performance gain) through execution plan tuning, indexing, database partitioning, and caching strategies; built **data profiling and monitoring** tools — log tables, audit tables, and traceability pipelines ensuring data quality and SLA compliance across high-traffic production schemas.
- **Translated complex business requirements into data-driven solutions**: analyzed AMI threshold distributions across 1.6M HUD/Tax Credit income records (5 years, 5 US states) to model eligibility patterns; built occupancy dashboards tracking recertification compliance rates, waitlist anomalies, and subsidy exposure — collaborated cross-functionally with compliance, operations, and product stakeholders to align data strategies with organizational goals.
- Owned **data governance and regulatory compliance** for affordable housing financial operations: automated income eligibility verification, AMI calculations, and recertification workflows under HUD, LIHTC, and HCV regulations; enforced **metadata standards and data privacy** requirements — zero compliance violations across 100K+ unit portfolios.
- **Achieved 100% PHI compliance** on AWS S3 migration with Libsodium-encrypted pipelines (encrypted 100% of resident data — phone numbers and SSNs); updated 50%+ of product codebase for **data privacy and regulatory compliance**; zero security incidents.
- Eliminated 80% of manual operational workload via **Python data pipeline automation**; shipped 20+ full-stack production modules across 120+ Agile sprints for 100K+ MAU at 99.5% uptime; **mentored 5+ junior analysts and engineers** in data analysis methods, SQL best practices, and quality management — conducted 300+ peer code reviews; maintained bug count below 10 for 6+ consecutive months.

Magizhs Tech Solution

May 2025 – Aug 2025

AI/ML Engineer Intern — Data Analysis · Compliance Automation · Data Quality Enforcement

- Built an agentic data pipeline (LangGraph stateful graph) applying **statistical decision logic and data-driven policy enforcement**: operations are profiled against regulatory standards, compliant outputs route to export, non-compliant ones route to rejection with cited, grounded reasons — enforcing **data governance** and service level requirements across multi-step workflows.

Entrata India (Db Xento Systems)

Jan 2017 – Sep 2018

Associate / Trainee Software Engineer — Data Pipelines · SQL · AWS S3 · Compliance Data

- Built affordable housing compliance data pipelines (PHP, PostgreSQL, JavaScript); designed SQL import scripts processing 1.6M HUD/Tax Credit income records; migrated thousands of client documents to AWS S3; redesigned document storage database architecture enforcing **data governance and metadata standards**; created 56 state-specific compliance documents with merge fields for housing eligibility verification.

PORTFOLIO PROJECTS

CrashClarity — Large-Scale Transportation Data Analysis & Predictive Modeling (104K Records)

Statistical Analysis · Data Visualization · EDA

2025 · Python, R, scikit-learn, Random Forest, Logistic Regression, Feature Engineering, Tableau, GitHub Pages · View Project

- Performed end-to-end **exploratory data analysis** on 104K+ UK transportation accident records — data profiling, cleansing, feature engineering across behavioral, environmental, and temporal variables; identified actionable patterns in road safety data aligned with operational improvement goals.
- Achieved **85%+ model accuracy** with Random Forest + Logistic Regression pipeline; deployed interactive **Tableau-style dashboard** on GitHub Pages for cross-functional stakeholder communication of data-driven insights and risk recommendations.

Leasel — Data Governance & Quality Enforcement Pipeline for Regulatory Compliance

Data Governance · Data Quality · Regulatory Compliance

2026 · Python, LangGraph (8-agent), FastAPI, TF-IDF, OCR, SQLite · github.com/venkatalakshmiparimala/Lease_complaince_rag

- Built a **data quality and governance pipeline** enforcing deterministic compliance rules on 1,240 residential lease documents (5 US states + federal, 1,050 synthetic records) — **93.6% precision** and 87.9% F1 across 10 compliance rules; 100% citation correctness; zero unsupported-claim rate.
- Designed **data monitoring and validation** layer (8-agent LangGraph graph) that profiles ingested documents for missing/conflicting clauses, flags data quality issues, and enforces **metadata standards** before any regulated eligibility or financial determination is made.

AI on FHIR — NLP-Driven Medical Records Query & HIPAA Data Governance

Data Governance · Regulatory Compliance · Data Privacy

2024 · Flask, LLMs, FHIR, React, Docker, spaCy, BERT, Embeddings · github.com/venkatalakshmiparimala/AI_on_FHIR

- Reduced manual clinical data retrieval time ~60% and enforced **HIPAA-compliant RBAC and data privacy** controls across structured medical records by building an NLP platform translating natural language queries to FHIR APIs — **regulatory compliance and data governance** embedded at the query layer; cross-functional stakeholder access tailored by role.

EDUCATION

M.S. Data Analytics Engineering — George Mason University, Virginia — May 2026 · GPA 3.87

Coursework: Statistical Analysis, Data Mining, Machine Learning, Data Visualization, Database Systems, Applied Analytics, Data Governance

B.Tech Computer Science & Engineering — JNTUA, India — 2013–2017

TECHNICAL SKILLS

Data Analysis & Analytics: Data Analysis, Exploratory Data Analysis, Statistical Analysis, Hypothesis Testing, Statistical Techniques, Statistical Modeling, Data Profiling, Data Cleansing, Data Monitoring, Business Intelligence, KPI Metrics, Operational Analytics, eCommerce Analytics

Data & Query: SQL (PostgreSQL, MySQL, Oracle, NoSQL), Distributed Data Stores, Apache Spark, PySpark, AWS Glue, Redshift, Python (Pandas, NumPy, scikit-learn), R, ETL, Data Pipelines

Visualization & Reporting: Tableau, Power BI, Data Visualization, Dashboard Design, Matplotlib, Seaborn, Plotly, ggplot2, Quarto, Stakeholder Reporting

Governance & Compliance: Data Governance, Metadata Standards, Regulatory Compliance, Data Privacy, Data Quality Management, Service Level Agreements (SLA), HIPAA Compliance, PHI Data Handling, Audit Trails

ML & Modeling: Random Forest, Logistic Regression, Isolation Forest, XGBoost, Feature Engineering, Model Evaluation, scikit-learn, PyTorch, NLP, BERT, spaCy, Anomaly Detection

Engineering & Infrastructure: Python, FastAPI, REST APIs, AWS (S3, EC2, RDS, Glue), Docker, CI/CD (GitHub Actions), PostgreSQL, MySQL, Microservices, LangChain, LangGraph, RAG Pipelines

Languages: Python (primary), SQL, R, PHP, JavaScript, TypeScript, PL/SQL, Bash